# Jaemin Eom – Curriculum Vitae

Computational Robot Design & Fabrication (CREATE) Lab.
Institute of Mechanical Engineering (IGM), School of Engineering
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## Research Interests

- Computational Design for Robotic Hands
- Physics-Based Simulation and Optimization
- Design and Control of Robotic Grippers
- Multi-Goal Path Planning for Robotic Manipulation
- Teleoperated Systems
- Tendon-Driven Mechanisms
- Soft Robotics

## Experience

Sep. 2025 - Present — **Postdoctoral Research Associate**
CREATE Lab, École Polytechnique Fédérale de Lausanne (EPFL), Switzerland
- Advisor: Prof. Josie Hughes

Mar. 2025 - Aug. 2025 — **Postdoctoral Research Associate**
Soft Robotics Research Center (SRRC), Biorobotics Lab, Seoul National University, Korea
- Advisor: Prof. Kyu-Jin Cho

## Education

Sep. 2017 - Feb. 2025 — **Ph.D. in Mechanical Engineering** (GPA : 3.77/4.3)
Seoul National University, Seoul, Korea
- Dissertation: Multi-Object Grasping Using Finger-to-Palm Translation for Pick-and-Place Tasks
- Advisor: Prof. Kyu-Jin Cho
- **Outstanding Doctoral Dissertation Award**

Mar. 2013 - Aug. 2017 — **B.S. in Mechanical Engineering** (GPA : 3.92/4.3)
Seoul National University, Seoul, Korea
- **Summa Cum Laude**

## PUBLICATIONS

### International Journal

1. **Jaemin Eom**, Sung Yol Yu, Woongbae Kim, Chunghoon Park, Kristine Yoonseo Lee, and Kyu-Jin Cho, "MOGrip: Gripper for multiobject grasping in pick-and-place tasks using translational movements of fingers," **Science Robotics (I.F. 26.1)**, vol. 9, eado3939, 2024. [Paper], [Video], [Project Page]
2. Yuna Yoo*, **Jaemin Eom***, MinJo Park, and Kyu-Jin Cho, "Compliant Suction Gripper with Seamless Deployment and Retraction for Robust Picking against Depth and Tilt Errors," **IEEE Robotics and Automation Letters (I.F. 4.6)**, vol.8, no.3, 2023. **(Co-first author)** [Paper], [Video], [Project Page]

3. Woongbae Kim, **Jaemin Eom**, and Kyu-Jin Cho, "A Dual-Origami Design that Enables the Quasisequential Deployment and Bending Motion of Soft Robots and Grippers," **Advanced Intelligent Systems (I.F. 6.8)**, vol. 4, no. 3, 2021. [Paper], [Video]
4. Jun-Young Lee, **Jaemin Eom**, Sung Yol Yu, and Kyu-Jin Cho, "Customization Methodology for Conformable Grasping Posture of Soft Grippers by Stiffness Patterning," **Frontiers in Robotics and AI**, vol. 7, 2020. [Paper]

### Referred Conference Paper

5. Jun-Young Lee, **Jaemin Eom**, Woo-Young Choi and Kyu-Jin Cho, "Soft LEGO: Bottom-up Design Platform for Soft Robotics," **2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)**, 2018, pp. 7513-7520. [Paper], [Video]

### Journals in Preparation

6. **Jaemin Eom,** and Kyu-Jin Cho, "Path Planning for Multi-Object Grasping in a Declutter Problem," in preparation. [Project Page]
7. **Jaemin Eom***, Haewoo Lee*, JaeHyun Lee, and Kyu-Jin Cho, "Robotic Hand Design for Multi-Object Grasping of Column-Shaped Objects," in preparation. **(Co-first author)** [Project Page]

### Patents

8. Kyu-Jin Cho, **Jaemin Eom**, Yuna Yoo, MinJo Park, "Longitudinal Deployable Vacuum Suction Cup," **PCT/KR2023/019352** (Patent Application, filed on Nov. 28th, 2023).
9. Kyu-Jin Cho, Jun-Young Lee, **Jaemin Eom**, "Soft Block Unit Comprising Expanding Block and Bending Block," **JP Patent 6620257** issued Nov. 22th, 2019.
10. Kyu-Jin Cho, Jaemin Eom, Yuna Yoo, MinJo Park, "Longitudinal Deployable Vacuum Suction Cup," **KR Patent 10-2624036** issued Jan. 8th, 2024.
11. Kyu-Jin Cho, **Jaemin Eom**, Sung Yol Yu, Woongbae Kim, "Gripper for Gripping Multi-Object with an Internal Storage," **KR Patent 10-2497956** issued Feb. 6th, 2023.
12. Kyu-Jin Cho, Jun-Young Lee, **Jaemin Eom**, "Soft Block Unit Comprising Expanding Block and Bending Block," **KR Patent 101950654** issued Apr. 14th, 2019.

### Dissertation

13. **Jaemin Eom**, "Multi-Object Grasping Using Finger-to-Palm Translation for Pick-and-Place Tasks," Seoul National University, Seoul, Korea. [pdf]

## Honor and Awards

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| Apr. 2026 | **Reviewer of the Year, *npj Robotics*** (Nature Portfolio), 2025 (Official announcement expected May 2026) |
| Feb. 2025 | **Outstanding Doctoral Dissertation Award** (Department of Mechanical Engineering) |
| Apr. 2021 | 1st prize winner, RoboSoft 2021 Manipulation Challenge, IEEE International Conference on Soft Robotics |
| Nov. 2020 | Silver Prize, 5th KSME-SEMES Open Innovation Challenge, Young Engineers Group |
| Dec. 2019 | Silver Prize, 4th KSME-SEMES Open Innovation Challenge, Young Engineers Group |
| Apr. 2019 | 3rd prize winner, RoboSoft 2019 Manipulation Challenge, IEEE International Conference on Soft Robotics |
| Feb. 2019 | **Bronze Prize, 25th SAMSUNG Humantech Paper Award** |
| Nov. 2023 | Honorable Mention, 8th KSME-SEMES Open Innovation Challenge (Young Engineers Group) |
| Nov. 2023 | Honorable Mention, 8th KSME-SEMES Open Innovation Challenge (Young Engineers Group) |

# Professional Service

### Peer Reviews (Total: 6)

- **Selected as "2025 Reviewer of the Year" by *npj Robotics***
- **Science Robotics: 1**
- **npj Robotics: 1**
- IEEE Robotics and Automation Letters: 2
- IEEE-RAS International Conference on Soft Robotics (RoboSoft): 1
- Intl. Conf. on Embodied Intelligence: 1

# Research Experience

Jan. 2026–Present
**Computational Design Optimization of a Robotic Hand for Tool Use**
- Optimized manufacturing-constrained hand parameters for robust tool use via Bayesian Optimization.
- Employed MuJoCo for efficient design space exploration and hardware feasibility validation.

Sep. 2025–Present
**LLM-Guided Task Planning for Heterogeneous Bimanual Manipulation**
- Leveraged LLMs for high-level task reasoning and distribution between asymmetric end-effectors.
- Validate cognitive reasoning and action chunking for complex tasks via teleoperation.

Sep. 2025–Present
**Integrated Hand System Development via Industry Collaboration**
- Integrated state-of-the-art sensors and motors into a robot hand via industry collaboration.
- Troubleshooting engineering issues during the integration phase.

Sep. 2024–Aug. 2025
**Manipulator path planning for multi-object grasping in declutter problem**
- Efficiently solved the decluttering problem by grasping and transporting multiple objects at once.
- Proposed an algorithm to find the minimum path for decluttering all given objects.

Mar. 2024-Aug. 2025
**Robotic hand design for multi-object grasping**
- Designed a robotic hand that sequentially grasps multiple objects, stores them in the palm, and transports them all at once.
- Analyzed finger links' length and joint stiffness for target motion through analytic model and simulation.
- Design the experiments and demonstrations.

Aug. 2019-Dec. 2024
**Multi-object grasping using finger-to-palm translation for pick-and-place tasks**
- Proposed finger-to-palm translation as a key manipulation skill for multi-object grasping in pick-and-place tasks.
- Presented a finger design enabling finger-to-palm translation.
- Introduced a soft conveyor palm design capable of storing multiple objects simultaneously.
- Designed the experiment, conducted experimental work, and performed demonstrations.
- Conducted analytic modeling and ABAQUS simulation.

Dec. 2020-Dec. 2022
**Compliant suction gripper with seamless deployment and retraction**
- Supervised a UROP student and submitted a paper to IEEE Robotics and Automation Letters.
- Designed a deployable body of a suction cup.
- Proposed a pneumatic circuit design for seamless deployment, picking, and retraction.
- Designed the experiments and demonstrations.

Jan. 2020 – Dec. 2022 **Development of a collaborative assistive robot arm utilizing foldable soft robot technology**
Funded by *Ministry of Trade, Industry & Energy*
- Integrated the developed foldable gripper and the developed robotic arm.

Nov. 2020- Dec. 2021 **Dual-Origami Design that Enables the Quasisequential Deployment and Bending Motion**
- Designed the experiments and demonstrations.

Jan. 2018 - Dec. 2020 **Development of modular gripper for small quantity production process**
Funded by *Korea Institute of Machinery & Materials*
- Principal investigator of research project.
- Controlled the developed soft gripper using ROS communication.
- Developed a customized soft gripper with task specific designs.

Sep. 2017 - May 2020 **Development of fundamental soft robotics technology for advanced soft grippers**
Funded by *Ministry of Trade, Industry & Energy*
- Principal investigator of research project.
- Developed pneumatically actuated soft gripper for various objects, especially e-commerce.
- Controlled the developed soft gripper using ROS communication.
- Benchmarked the Amazon Picking Challenge to analyze feasibility of gripper.

Jan. 2018 – Dec. 2019 **A hybrid gripper with pinching and suction grasp modes using a soft reconfigurable structure**
- Designed of a soft reconfigurable structure that functions as a gripper with fingers and transforms into a suction cup upon everting.
- Proposed a tendon routing for adaptive grasping.

Jan. 2017 – Dec. 2018 **Soft LEGO: Bottom-up design platform for soft robotics**
- Proposed modular soft pneumatic actuator design compatible with LEGO.
- Utilized the ABAQUS simulation to predict the behavior of a soft pneumatic module.
- Designed the demonstrations.

# Teaching Experience

Sep. 2019 - Dec. 2019 **Teaching Assistant**
**Dynamics (M2794.001200)**
Seoul National University
*Supervisor : Prof. Kyu-Jin Cho*

Mar. 2018 - Jun. 2018 **Teaching Assistant**
**Management in Mechanical Engineering 1 (M2794.004500)**
Seoul National University
*Supervisor : Prof. Young-sang Yoo*

Feb. 2026 - Present **M.S. Thesis/Semester Project Tutoring** (Prof. Josie Hughes)
Supervised **one** M.S. student for thesis research
Mentored **one** M.S. student for semester project

Jan. 2025 – Aug. 2026
Feb. 2023 - Dec. 2024
Dec. 2020 - Dec. 2022
Sep. 2018 - Aug. 2019
**B.S Thesis/UROP Tutoring** (Prof. Kyu-Jin Cho)
Supervised the B.S. Thesis of **one** undergraduate student
Mentored **six** students for the Undergraduate Research Opportunities

# Technical Skills

**Design & Manufacturing, Simulation, Motion Planning, Embedded system, Control**

1. Actuator System Design and Control (Tendon-driven systems for the under-actuated gripper, Pneumatic circuit design, Low-level control)
2. Physics-Based Simulation and Analysis (MuJoCo, ABAQUS, MATLAB)
3. Motion Planning for Robotic Manipulator
4. Mechanism Design and Prototyping (Robotic hand, MOGrip, Deployable suction gripper, Experimental setups)
5. CAD design (SOLIDWORKS)
6. Manufacturing (Elastomer molding for soft robot fabrication, Laser cutting, 3D printing, Heat press)
7. Circuit design (Eagle CAD)